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ENCLOSURE ASSEMBLY FOR HOUSING A BANK NOTES ACCEPTOR WITHIN AN ELECTRONIC GAMING MACHINE

Abstract

An enclosure assembly for securely housing a bank notes acceptor (BNA) for use with an electronic gaming machine (EGM) is provided. The enclosure assembly includes a housing defining an interior for receiving the BNA. The housing includes an engagement bracket extending between a first end coupled to a side wall and a second, free end. The door assembly includes a door panel and a lock assembly. The lock assembly includes a lock frame including a side wall and a projection provided thereon, the projection extends from the side wall and towards the engagement bracket. The lock assembly further includes a prong latch coupled to the lock frame that is engageable with the engagement bracket to lock the door assembly. The engagement bracket is resiliently flexible and is configured to move into contact with the projection to enhance a structural integrity of the enclosure assembly.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/581,950, filed Feb. 20, 2024, which is a continuation of U.S. patent application Ser. No. 17/831,246, filed Jun. 2, 2022, and issued as U.S. Pat. No. 11,941,934, which claims priority to U.S. Provisional Application No. 63/197,838, filed Jun. 7, 2021, the contents and disclosure of which are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

[0002] The field of relates generally to an enclosure assembly that houses a bank notes acceptor and, more specifically, an enclosure assembly having a torque damping door assembly, in which the door assembly engages with the housing to enhance a structural integrity of the enclosure assembly for added security within an electronic gaming machine.

BACKGROUND

[0003] Electronic gaming machines (“EGMs”) or gaming devices provide a variety of wagering games such as slot games, video poker games, video blackjack games, roulette games, video bingo games, keno games and other types of games that are frequently offered at casinos and other locations. Play on EGMs typically involves a player establishing a credit balance by inputting money, or another form of monetary credit, and placing a monetary wager (from the credit balance) on one or more outcomes of an instance (or single play) of a primary or base game. In some cases, a player may qualify for a special mode of the base game, a secondary game, or a bonus round of the base game by attaining a certain winning combination or triggering event in, or related to, the base game, or after the player is randomly awarded the special mode, secondary game, or bonus round. In the special mode, secondary game, or bonus round, the player is given an opportunity to win extra game credits, game tokens or other forms of payout. In the case of “game credits” that are awarded during play, the game credits are typically added to a credit meter total on the EGM and can be provided to the player upon completion of a gaming session or when the player wants to “cash out.”

[0004] “Slot” type games are often displayed to the player in the form of various symbols arrayed in a row-by-column grid or matrix. Specific matching combinations of symbols along predetermined paths (or paylines) through the matrix indicate the outcome of the game. The display typically highlights winning combinations/outcomes for identification by the player. Matching combinations and their corresponding awards are usually shown in a “pay-table” which is available to the player for reference. Often, the player may vary his/her wager to include differing numbers of paylines and/or the amount bet on each line. By varying the wager, the player may sometimes alter the frequency or number of winning combinations, frequency or number of secondary games, and/or the amount awarded.

[0005] Typical games use a random number generator (RNG) to randomly determine the outcome of each game. The game is designed to return a certain percentage of the amount wagered back to the player over the course of many plays or instances of the game, which is generally referred to as return to player (RTP). The RTP and randomness of the RNG ensure the fairness of the games and

are highly regulated. Upon initiation of play, the RNG randomly determines a game outcome and symbols are then selected which correspond to that outcome. Notably, some games may include an element of skill on the part of the player and are therefore not entirely random.

BRIEF DESCRIPTION

[0006] In one aspect, an enclosure assembly for securely housing a bank notes acceptor (BNA) for use with an EGM is provided. The enclosure assembly includes a housing having a side wall at least partially defining an interior sized and shaped to receive the BNA. The housing further includes an engagement bracket extending between a first end coupled to the side wall and a second, free end. The enclosure assembly further includes a door assembly hingeably coupled to the housing and moveable between an open position and a closed position. The door assembly includes a door panel and a lock assembly coupled to the door panel for selectively locking the door panel on the housing. The lock assembly includes a lock frame including a side wall and a projection provided on the side wall, the projection extending outward from the side wall and towards the engagement bracket when the door assembly is in the closed position. The lock assembly further includes a prong latch moveably coupled to the lock frame. The prong latch is selectively engageable with the engagement bracket to lock the door assembly in the closed position. The engagement bracket is resiliently flexible and, in response to an opening force being applied on the door panel when the door assembly is locked in the closed position, the engagement bracket is configured to move into contact with the projection to enhance a structural integrity of the enclosure assembly.

[0007] In another aspect, an EGM is provided. The EGM includes a cabinet, a display coupled to the cabinet for display an electronic game thereon, and an enclosure assembly for securely housing a bank notes acceptor (BNA) therein. The enclosure assembly includes a housing including a side wall at least partially defining an interior sized and shaped to receive the BNA. The housing further includes an engagement bracket extending between a first end coupled to the side wall and a second, free end. The enclosure assembly further includes a door panel hingeably coupled to the housing and moveable between an open position and a closed position and a lock frame coupled to the door panel. The lock frame includes a side wall and a projection provided on the side wall, the projection extending outward from the side wall and towards the engagement bracket when the door panel is in the closed position. The enclosure assembly also includes a prong latch moveably coupled to the lock frame for selectively locking the door panel on the housing, the prong latch being selectively engageable with the engagement bracket to lock the door panel in the closed position. When the door panel is locked in the closed position, the engagement bracket is resiliently flexible to move into contact with the projection in response to an opening force being applied on the door panel to enhance a structural integrity of the enclosure assembly.

[0008] In yet another aspect, a method of securing a bank notes acceptor (BNA) within an enclosure assembly of an EGM is provided. The method includes providing a housing including a side wall at least partially defining an interior sized and shaped to receive the BNA, the housing further including an engagement bracket extending between a first end coupled to the side wall and a second, free end. The method further includes moving a door assembly from an open position to a closed position, the door assembly being hingedly coupled to the housing and including a door panel and a lock frame coupled to the door panel, the lock frame including a side wall and a projection provided on the side wall and extending outward therefrom towards the engagement bracket when the door assembly is in the closed position. The method further includes moving a prong latch into engagement with the engagement bracket to lock the door assembly in the closed position, the prong latch being moveably coupled to the lock frame. The engagement bracket is resiliently flexible and, in response to an opening force being applied on the door panel when the door assembly is locked in the closed position, the engagement bracket is configured to move into contact with the projection to enhance a structural integrity of the enclosure assembly.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] An example embodiment of the subject matter disclosed will now be described with reference to the accompanying drawings.

[0010] FIG. 1 is an exemplary diagram showing several EGMs networked with various gaming related servers.

[0011] FIG. 2A is a block diagram showing various functional elements of an exemplary EGM.

[0012] FIG. 2B depicts a casino gaming environment according to one example.

[0013] FIG. 2C is a diagram that shows examples of components of a system for providing online gaming according to some aspects of the present disclosure.

[0014] FIG. 3 illustrates, in block diagram form, an implementation of a game processing architecture algorithm that implements a game processing pipeline for the play of a game in accordance with various implementations described herein.

[0015] FIG. 4A is a perspective view of a portion of an EGM that includes an enclosure assembly for securely housing a currency acceptor, such as a bank notes acceptor (BNA), a bill validator, and the like.

[0016] FIG. 4B is a perspective view of the enclosure assembly shown in FIG. 4A.

[0017] FIG. 5 is a front view of the enclosure assembly shown in FIG. 4B.

[0018] FIG. 6 is an exploded view of the enclosure assembly shown in FIG. 4B.

[0019] FIG. 7 is a perspective view of a torque damper assembly of the enclosure assembly shown in FIG. 4B.

[0020] FIG. 8 is a perspective view of a portion of the enclosure assembly shown in FIG. 4A, with a door panel of the enclosure assembly removed to reveal a lock assembly.

[0021] FIG. 9 is a perspective view of a first side of the lock assembly shown in FIG. 8.

[0022] FIG. 10 is a perspective view of a second side of the lock assembly shown in FIG. 8.

[0023] FIG. 11 is a top view of the second side of the lock assembly shown in FIG. 10.

[0024] FIG. 12 is another perspective view of a portion of the enclosure assembly shown in FIG. 4A, with the door panel removed.

[0025] FIG. 13 is another perspective view of the enclosure assembly shown in FIG. 4A.

[0026] FIG. 14 is a side view of a portion of the enclosure assembly shown in FIG. 4A, showing the door panel in a closed position and a gap closing assembly in a raised position.

[0027] FIG. 15 is a perspective view of the portion of the enclosure assembly shown in FIG. 14, showing the door panel in a partially opened position and the gap closing assembly in a lowered position.

DETAILED DESCRIPTION

[0028] An enclosure assembly for securely housing a bank notes acceptor (BNA) includes a housing defining an opening sized and shaped to receive the BNA. The housing includes an exterior portion and an interior portion. The housing includes an engagement bracket having a free end. A door is hingedly coupled to the housing and is moveable between an open position and a closed position and includes a lock assembly thereon. The lock assembly includes a lock frame having a projection and a prong latch moveably coupled to the lock frame. When the door is closed, the lock assembly may lock the door by moving the prong latch into engagement with the engagement bracket. The engagement bracket is resiliently flexible and may move or bend, in response to an opening force being applied on the door assembly is locked in the closed position, to contact the projection and restrain against opening of the door. The engagement bracket and lock assembly provide a compact mechanism for securely locking the door. For example, because the engagement bracket is resiliently flexible, the BNA may be easily removed from the housing without being obstructed by the engagement bracket.

[0029] FIG. 1 illustrates several different models of EGMs which may be networked to various gaming related servers. Shown is a system **100** in a gaming environment including one or more server computers **102** (e.g., slot servers of a casino) that are in communication, via a communications network, with one or more gaming devices **104A-104X** (EGMs, slots, video poker, bingo machines, etc.) that can implement one or more aspects of the present disclosure. The gaming devices **104A-104X** may alternatively be portable and/or remote gaming devices such as, but not limited to, a smart phone, a tablet, a laptop, or a game console. Gaming devices **104A-104X** utilize specialized software and/or hardware to form non-generic, particular machines or apparatuses that comply with regulatory requirements regarding devices used for wagering or games of chance that provide monetary awards.

[0030] Communication between the gaming devices **104A-104X** and the server computers **102**, and among the gaming devices **104A-104X**, may be direct or indirect using one or more communication protocols. As an example, gaming devices **104A-104X** and the server computers **102** can communicate over one or more communication networks, such as over the Internet through a website maintained by a computer on a remote server or over an online data network including commercial online service providers, Internet service providers, private networks (e.g., local area networks and enterprise networks), and the like (e.g., wide area networks). The communication networks could allow gaming devices **104A-104X** to communicate with one another and/or the server computers **102** using a variety of communication-based technologies, such as radio frequency (RF) (e.g., wireless fidelity (WiFi®) and Bluetooth®), cable TV, satellite links and the like.

[0031] In some implementation, server computers **102** may not be necessary and/or preferred. For example, in one or more implementations, a stand-alone gaming device such as gaming device **104A**, gaming device **104B** or any of the other gaming devices **104C-104X** can implement one or more aspects of the present disclosure. However, it is typical to find multiple EGMs connected to networks implemented with one or more of the different server computers **102** described herein.

[0032] The server computers **102** may include a central determination gaming system server **106**, a ticket-in-ticket-out (TITO) system server **108**, a player tracking system server **110**, a progressive system server **112**, and/or a casino management system server **114**. Gaming devices **104A-104X** may include features to enable operation of any or all servers for use by the player and/or operator (e.g., the casino, resort, gaming establishment, tavern, pub, etc.). For example, game outcomes may be generated on a central determination gaming system server **106** and then transmitted over the network to any of a group of remote terminals or remote gaming devices **104A-104X** that utilize the game outcomes and display the results to the players.

[0033] Gaming device **104A** is often of a cabinet construction which may be aligned in rows or banks of similar devices for placement and operation on a casino floor. The gaming device **104A** often includes a main door which provides access to the interior of the cabinet. Gaming device **104A** typically includes a button area or button deck **120** accessible by a player that is configured with input switches or buttons **122**, an access channel for a bill validator **124**, and/or an access channel for a ticket-out printer **126**.

[0034] In FIG. 1, gaming device **104A** is shown as a Relm XL™ model gaming device manufactured by Aristocrat® Technologies, Inc. As shown, gaming device **104A** is a reel machine having a gaming display area **118** comprising a number (typically 3 or 5) of mechanical reels **130** with various symbols displayed on them. The mechanical reels **130** are independently spun and stopped to show a set of symbols within the gaming display area **118** which may be used to determine an outcome to the game.

[0035] In many configurations, the gaming device **104A** may have a main display **128** (e.g., video display monitor) mounted to, or above, the gaming display area **118**. The main display **128** can be a high-resolution liquid crystal display (LCD), plasma, light emitting diode (LED), or organic light emitting diode (OLED) panel which may be flat or curved as shown, a cathode ray tube, or other

conventional electronically controlled video monitor.

[0036] In some implementations, the bill validator **124** may also function as a “ticket-in” reader that allows the player to use a casino issued credit ticket to load credits onto the gaming device **104A** (e.g., in a cashless ticket (“TITO”) system). In such cashless implementations, the gaming device **104A** may also include a “ticket-out” printer **126** for outputting a credit ticket when a “cash out” button is pressed. Cashless TITO systems are used to generate and track unique bar-codes or other indicators printed on tickets to allow players to avoid the use of bills and coins by loading credits using a ticket reader and cashing out credits using a ticket-out printer **126** on the gaming device **104A**. The gaming device **104A** can have hardware meters for purposes including ensuring regulatory compliance and monitoring the player credit balance. In addition, there can be additional meters that record the total amount of money wagered on the gaming device, total amount of money deposited, total amount of money withdrawn, total amount of winnings on gaming device **104A**.

[0037] In some implementations, a player tracking card reader **144**, a transceiver for wireless communication with a mobile device (e.g., a player's smartphone), a keypad **146**, and/or an illuminated display **148** for reading, receiving, entering, and/or displaying player tracking information is provided in gaming device **104A**. In such implementations, a game controller within the gaming device **104A** can communicate with the player tracking system server **110** to send and receive player tracking information.

[0038] Gaming device **104A** may also include a bonus topper wheel **134**. When bonus play is triggered (e.g., by a player achieving a particular outcome or set of outcomes in the primary game), bonus topper wheel **134** is operative to spin and stop with indicator arrow **136** indicating the outcome of the bonus game. Bonus topper wheel **134** is typically used to play a bonus game, but it could also be incorporated into play of the base or primary game.

[0039] A candle **138** may be mounted on the top of gaming device **104A** and may be activated by a player (e.g., using a switch or one of buttons **122**) to indicate to operations staff that gaming device **104A** has experienced a malfunction or the player requires service. The candle **138** is also often used to indicate a jackpot has been won and to alert staff that a hand payout of an award may be needed.

[0040] There may also be one or more information panels **152** which may be a back-lit, silkscreened glass panel with lettering to indicate general game information including, for example, a game denomination (e.g., \$0.25 or \$1), pay lines, pay tables, and/or various game related graphics. In some implementations, the information panel(s) **152** may be implemented as an additional video display.

[0041] Gaming devices **104A** have traditionally also included a handle **132** typically mounted to the side of main cabinet **116** which may be used to initiate game play.

[0042] Many or all the above described components can be controlled by circuitry (e.g., a game controller) housed inside the main cabinet **116** of the gaming device **104A**, the details of which are shown in FIG. 2A.

[0043] An alternative example gaming device **104B** illustrated in FIG. 1 is the Arc™ model gaming device manufactured by Aristocrat® Technologies, Inc. Note that where possible, reference numerals identifying similar features of the gaming device **104A** implementation are also identified in the gaming device **104B** implementation using the same reference numbers. Gaming device **104B** does not include physical reels and instead shows game play functions on main display **128**. An optional topper screen **140** may be used as a secondary game display for bonus play, to show game features or attraction activities while a game is not in play, or any other information or media desired by the game designer or operator. In some implementations, the optional topper screen **140** may also or alternatively be used to display progressive jackpot prizes available to a player during play of gaming device **104B**.

[0044] Example gaming device **104B** includes a main cabinet **116** including a main door which

opens to provide access to the interior of the gaming device **104B**. The main or service door is typically used by service personnel to refill the ticket-out printer **126** and collect bills and tickets inserted into the bill validator **124**. The main or service door may also be accessed to reset the machine, verify and/or upgrade the software, and for general maintenance operations.

[0045] Another example gaming device **104C** shown is the Helix™ model gaming device manufactured by Aristocrat® Technologies, Inc. Gaming device **104C** includes a main display **128A** that is in a landscape orientation. Although not illustrated by the front view provided, the main display **128A** may have a curvature radius from top to bottom, or alternatively from side to side. In some implementations, main display **128A** is a flat panel display. Main display **128A** is typically used for primary game play while secondary display **128B** is typically used for bonus game play, to show game features or attraction activities while the game is not in play or any other information or media desired by the game designer or operator. In some implementations, example gaming device **104C** may also include speakers **142** to output various audio such as game sound, background music, etc.

[0046] Many different types of games, including mechanical slot games, video slot games, video poker, video black jack, video pachinko, keno, bingo, and lottery, may be provided with or implemented within the depicted gaming devices **104A-104C** and other similar gaming devices. Each gaming device may also be operable to provide many different games. Games may be differentiated according to themes, sounds, graphics, type of game (e.g., slot game vs. card game vs. game with aspects of skill), denomination, number of paylines, maximum jackpot, progressive or non-progressive, bonus games, and may be deployed for operation in Class 2 or Class 3, etc.

[0047] FIG. 2A is a block diagram depicting exemplary internal electronic components of a gaming device **200** connected to various external systems. All or parts of the gaming device **200** shown could be used to implement any one of the example gaming devices **104A-X** depicted in FIG. 1. As shown in FIG. 2A, gaming device **200** includes a topper display **216** or another form of a top box (e.g., a topper wheel, a topper screen, etc.) that sits above cabinet **218**. Cabinet **218** or topper display **216** may also house a number of other components which may be used to add features to a game being played on gaming device **200**, including speakers **220**, a ticket printer **222** which prints bar-coded tickets or other media or mechanisms for storing or indicating a player's credit value, a ticket reader **224** which reads bar-coded tickets or other media or mechanisms for storing or indicating a player's credit value, and a player tracking interface **232**. Player tracking interface **232** may include a keypad **226** for entering information, a player tracking display **228** for displaying information (e.g., an illuminated or video display), a card reader **230** for receiving data and/or communicating information to and from media or a device such as a smart phone enabling player tracking. FIG. 2 also depicts utilizing a ticket printer **222** to print tickets for a TITO system server **108**. Gaming device **200** may further include a bill validator **234**, player-input buttons **236** for player input, cabinet security sensors **238** to detect unauthorized opening of the cabinet **218**, a primary game display **240**, and a secondary game display **242**, each coupled to and operable under the control of game controller **202**.

[0048] The games available for play on the gaming device **200** are controlled by a game controller **202** that includes one or more processors **204**. Processor **204** represents a general-purpose processor, a specialized processor intended to perform certain functional tasks, or a combination thereof. As an example, processor **204** can be a central processing unit (CPU) that has one or more multi-core processing units and memory mediums (e.g., cache memory) that function as buffers and/or temporary storage for data. Alternatively, processor **204** can be a specialized processor, such as an application specific integrated circuit (ASIC), graphics processing unit (GPU), field-programmable gate array (FPGA), digital signal processor (DSP), or another type of hardware accelerator. In another example, processor **204** is a system on chip (SoC) that combines and integrates one or more general-purpose processors and/or one or more specialized processors. Although FIG. 2A illustrates that game controller **202** includes a single processor **204**, game

controller **202** is not limited to this representation and instead can include multiple processors **204** (e.g., two or more processors).

[0049] FIG. 2A illustrates that processor **204** is operatively coupled to memory **208**. Memory **208** is defined herein as including volatile and nonvolatile memory and other types of non-transitory data storage components. Volatile memory is memory that do not retain data values upon loss of power. Nonvolatile memory is memory that do retain data upon a loss of power. Examples of memory **208** include random access memory (RAM), read-only memory (ROM), hard disk drives, solid-state drives, universal serial bus (USB) flash drives, memory cards accessed via a memory card reader, floppy disks accessed via an associated floppy disk drive, optical discs accessed via an optical disc drive, magnetic tapes accessed via an appropriate tape drive, and/or other memory components, or a combination of any two or more of these memory components. In addition, examples of RAM include static random access memory (SRAM), dynamic random access memory (DRAM), magnetic random access memory (MRAM), and other such devices. Examples of ROM include a programmable read-only memory (PROM), an erasable programmable read-only memory (EPROM), an electrically erasable programmable read-only memory (EEPROM), or other like memory device. Even though FIG. 2A illustrates that game controller **202** includes a single memory **208**, game controller **202** could include multiple memories **208** for storing program instructions and/or data.

[0050] Memory **208** can store one or more game programs **206** that provide program instructions and/or data for carrying out various implementations (e.g., game mechanics) described herein. Stated another way, game program **206** represents an executable program stored in any portion or component of memory **208**. In one or more implementations, game program **206** is embodied in the form of source code that includes human-readable statements written in a programming language or machine code that contains numerical instructions recognizable by a suitable execution system, such as a processor **204** in a game controller or other system. Examples of executable programs include: (1) a compiled program that can be translated into machine code in a format that can be loaded into a random access portion of memory **208** and run by processor **204**; (2) source code that may be expressed in proper format such as object code that is capable of being loaded into a random access portion of memory **208** and executed by processor **204**; and (3) source code that may be interpreted by another executable program to generate instructions in a random access portion of memory **208** to be executed by processor **204**.

[0051] Alternatively, game programs **206** can be set up to generate one or more game instances based on instructions and/or data that gaming device **200** exchanges with one or more remote gaming devices, such as a central determination gaming system server **106** (not shown in FIG. 2A but shown in FIG. 1). For purpose of this disclosure, the term “game instance” refers to a play or a round of a game that gaming device **200** presents (e.g., via a user interface (UI)) to a player. The game instance is communicated to gaming device **200** via the network **214** and then displayed on gaming device **200**. For example, gaming device **200** may execute game program **206** as video streaming software that allows the game to be displayed on gaming device **200**. When a game is stored on gaming device **200**, it may be loaded from memory **208** (e.g., from a read only memory (ROM)) or from the central determination gaming system server **106** to memory **208**.

[0052] Gaming devices, such as gaming device **200**, are highly regulated to ensure fairness and, in many cases, gaming device **200** is operable to award monetary awards (e.g., typically dispensed in the form of a redeemable voucher). Therefore, to satisfy security and regulatory requirements in a gaming environment, hardware and software architectures are implemented in gaming devices **200** that differ significantly from those of general-purpose computers. Adapting general purpose computers to function as gaming devices **200** is not simple or straightforward because of: (1) the regulatory requirements for gaming devices **200**, (2) the harsh environment in which gaming devices **200** operate, (3) security requirements, (4) fault tolerance requirements, and (5) the requirement for additional special purpose componentry enabling functionality of an EGM. These

differences require substantial engineering effort with respect to game design implementation, game mechanics, hardware components, and software.

[0053] One regulatory requirement for games running on gaming device **200** generally involves complying with a certain level of randomness. Typically, gaming jurisdictions mandate that gaming devices **200** satisfy a minimum level of randomness without specifying how a gaming device **200** should achieve this level of randomness. To comply, FIG. 2A illustrates that gaming device **200** could include an RNG **212** that utilizes hardware and/or software to generate RNG outcomes that lack any pattern. The RNG operations are often specialized and non-generic in order to comply with regulatory and gaming requirements. For example, in a slot game, game program **206** can initiate multiple RNG calls to RNG **212** to generate RNG outcomes, where each RNG call and RNG outcome corresponds to an outcome for a reel. In another example, gaming device **200** can be a Class II gaming device where RNG **212** generates RNG outcomes for creating Bingo cards. In one or more implementations, RNG **212** could be one of a set of RNGs operating on gaming device **200**. More generally, an output of the RNG **212** can be the basis on which game outcomes are determined by the game controller **202**. Game developers could vary the degree of true randomness for each RNG (e.g., pseudorandom) and utilize specific RNGs depending on game requirements. The output of the RNG **212** can include a random number or pseudorandom number (either is generally referred to as a “random number”).

[0054] In FIG. 2A, RNG **212** and hardware RNG **244** are shown in dashed lines to illustrate that RNG **212**, hardware RNG **244**, or both can be included in gaming device **200**. In one implementation, instead of including RNG **212**, gaming device **200** could include a hardware RNG **244** that generates RNG outcomes. Analogous to RNG **212**, hardware RNG **244** performs specialized and non-generic operations in order to comply with regulatory and gaming requirements. For example, because of regulation requirements, hardware RNG **244** could be a random number generator that securely produces random numbers for cryptography use. The gaming device **200** then uses the secure random numbers to generate game outcomes for one or more game features. In another implementation, the gaming device **200** could include both hardware RNG **244** and RNG **212**. RNG **212** may utilize the RNG outcomes from hardware RNG **244** as one of many sources of entropy for generating secure random numbers for the game features.

[0055] Another regulatory requirement for running games on gaming device **200** includes ensuring a certain level of RTP. Similar to the randomness requirement discussed above, numerous gaming jurisdictions also mandate that gaming device **200** provides a minimum level of RTP (e.g., RTP of at least 75%). A game can use one or more lookup tables (also called weighted tables) as part of a technical solution that satisfies regulatory requirements for randomness and RTP. In particular, a lookup table can integrate game features (e.g., trigger events for special modes or bonus games; newly introduced game elements such as extra reels, new symbols, or new cards; stop positions for dynamic game elements such as spinning reels, spinning wheels, or shifting reels; or card selections from a deck) with random numbers generated by one or more RNGs, so as to achieve a given level of volatility for a target level of RTP. (In general, volatility refers to the frequency or probability of an event such as a special mode, payout, etc. For example, for a target level of RTP, a higher-volatility game may have a lower payout most of the time with an occasional bonus having a very high payout, while a lower-volatility game has a steadier payout with more frequent bonuses of smaller amounts.) Configuring a lookup table can involve engineering decisions with respect to how RNG outcomes are mapped to game outcomes for a given game feature, while still satisfying regulatory requirements for RTP. Configuring a lookup table can also involve engineering decisions about whether different game features are combined in a given entry of the lookup table or split between different entries (for the respective game features), while still satisfying regulatory requirements for RTP and allowing for varying levels of game volatility.

[0056] FIG. 2A illustrates that gaming device **200** includes an RNG conversion engine **210** that

translates the RNG outcome from RNG **212** to a game outcome presented to a player. To meet a designated RTP, a game developer can set up the RNG conversion engine **210** to utilize one or more lookup tables to translate the RNG outcome to a symbol element, stop position on a reel strip layout, and/or randomly chosen aspect of a game feature. As an example, the lookup tables can regulate a prize payout amount for each RNG outcome and how often the gaming device **200** pays out the prize payout amounts. The RNG conversion engine **210** could utilize one lookup table to map the RNG outcome to a game outcome displayed to a player and a second lookup table as a pay table for determining the prize payout amount for each game outcome. The mapping between the RNG outcome to the game outcome controls the frequency in hitting certain prize payout amounts. [0057] FIG. 2A also depicts that gaming device **200** is connected over network **214** to player tracking system server **110**. Player tracking system server **110** may be, for example, an OASIS® system manufactured by Aristocrat® Technologies, Inc. Player tracking system server **110** is used to track play (e.g. amount wagered, games played, time of play and/or other quantitative or qualitative measures) for individual players so that an operator may reward players in a loyalty program. The player may use the player tracking interface **232** to access his/her account information, activate free play, and/or request various information. Player tracking or loyalty programs seek to reward players for their play and help build brand loyalty to the gaming establishment. The rewards typically correspond to the player's level of patronage (e.g., to the player's playing frequency and/or total amount of game plays at a given casino). Player tracking rewards may be complimentary and/or discounted meals, lodging, entertainment and/or additional play. Player tracking information may be combined with other information that is now readily obtainable by a casino management system.

[0058] When a player wishes to play the gaming device **200**, he/she can insert cash or a ticket voucher through a coin acceptor (not shown) or bill validator **234** to establish a credit balance on the gaming device. The credit balance is used by the player to place wagers on instances of the game and to receive credit awards based on the outcome of winning instances. The credit balance is decreased by the amount of each wager and increased upon a win. The player can add additional credits to the balance at any time. The player may also optionally insert a loyalty club card into the card reader **230**. During the game, the player views with one or more UIs, the game outcome on one or more of the primary game display **240** and secondary game display **242**. Other game and prize information may also be displayed.

[0059] For each game instance, a player may make selections, which may affect play of the game. For example, the player may vary the total amount wagered by selecting the amount bet per line and the number of lines played. In many games, the player is asked to initiate or select options during course of game play (such as spinning a wheel to begin a bonus round or select various items during a feature game). The player may make these selections using the player-input buttons **236**, the primary game display **240** which may be a touch screen, or using some other device which enables a player to input information into the gaming device **200**.

[0060] During certain game events, the gaming device **200** may display visual and auditory effects that can be perceived by the player. These effects add to the excitement of a game, which makes a player more likely to enjoy the playing experience. Auditory effects include various sounds that are projected by the speakers **220**. Visual effects include flashing lights, strobing lights or other patterns displayed from lights on the gaming device **200** or from lights behind the information panel **152** (FIG. 1).

[0061] When the player is done, he/she cashes out the credit balance (typically by pressing a cash out button to receive a ticket from the ticket printer **222**). The ticket may be “cashed-in” for money or inserted into another machine to establish a credit balance for play.

[0062] Additionally, or alternatively, gaming devices **104A-104X** and **200** can include or be coupled to one or more wireless transmitters, receivers, and/or transceivers (not shown in FIGS. 1 and 2A) that communicate (e.g., Bluetooth® or other near-field communication technology) with

one or more mobile devices to perform a variety of wireless operations in a casino environment. Examples of wireless operations in a casino environment include detecting the presence of mobile devices, performing credit, points, comps, or other marketing or hard currency transfers, establishing wagering sessions, and/or providing a personalized casino-based experience using a mobile application. In one implementation, to perform these wireless operations, a wireless transmitter or transceiver initiates a secure wireless connection between a gaming device **104A-104X** and **200** and a mobile device. After establishing a secure wireless connection between the gaming device **104A-104X** and **200** and the mobile device, the wireless transmitter or transceiver does not send and/or receive application data to and/or from the mobile device. Rather, the mobile device communicates with gaming devices **104A-104X** and **200** using another wireless connection (e.g., WiFi® or cellular network). In another implementation, a wireless transceiver establishes a secure connection to directly communicate with the mobile device. The mobile device and gaming device **104A-104X** and **200** sends and receives data utilizing the wireless transceiver instead of utilizing an external network. For example, the mobile device would perform digital wallet transactions by directly communicating with the wireless transceiver. In one or more implementations, a wireless transmitter could broadcast data received by one or more mobile devices without establishing a pairing connection with the mobile devices.

[0063] Although FIGS. **1** and **2A** illustrate specific implementations of a gaming device (e.g., gaming devices **104A-104X** and **200**), the disclosure is not limited to those implementations shown in FIGS. **1** and **2**. For example, not all gaming devices suitable for implementing implementations of the present disclosure necessarily include top wheels, top boxes, information panels, cashless ticket systems, and/or player tracking systems. Further, some suitable gaming devices have only a single game display that includes only a mechanical set of reels and/or a video display, while others are designed for bar counters or tabletops and have displays that face upwards. Gaming devices **104A-104X** and **200** may also include other processors that are not separately shown. Using FIG. **2A** as an example, gaming device **200** could include display controllers (not shown in FIG. **2A**) configured to receive video input signals or instructions to display images on game displays **240** and **242**. Alternatively, such display controllers may be integrated into the game controller **202**. The use and discussion of FIGS. **1** and **2** are examples to facilitate ease of description and explanation.

[0064] FIG. **2B** depicts a casino gaming environment according to one example. In this example, the casino **251** includes banks **252** of EGMs **104**. In this example, each bank **252** of EGMs **104** includes a corresponding gaming signage system **254** (also shown in FIG. **2A**). According to this implementation, the casino **251** also includes mobile gaming devices **256**, which are also configured to present wagering games in this example. The mobile gaming devices **256** may, for example, include tablet devices, cellular phones, smart phones and/or other handheld devices. In this example, the mobile gaming devices **256** are configured for communication with one or more other devices in the casino **251**, including but not limited to one or more of the server computers **102**, via wireless access points **258**.

[0065] According to some examples, the mobile gaming devices **256** may be configured for stand-alone determination of game outcomes. However, in some alternative implementations the mobile gaming devices **256** may be configured to receive game outcomes from another device, such as the central determination gaming system server **106**, one of the EGMs **104**, etc.

[0066] Some mobile gaming devices **256** may be configured to accept monetary credits from a credit or debit card, via a wireless interface (e.g., via a wireless payment app), via tickets, via a patron casino account, etc. However, some mobile gaming devices **256** may not be configured to accept monetary credits via a credit or debit card. Some mobile gaming devices **256** may include a ticket reader and/or a ticket printer whereas some mobile gaming devices **256** may not, depending on the particular implementation.

[0067] In some implementations, the casino **251** may include one or more kiosks **260** that are configured to facilitate monetary transactions involving the mobile gaming devices **256**, which may

include cash out and/or cash in transactions. The kiosks **260** may be configured for wired and/or wireless communication with the mobile gaming devices **256**. The kiosks **260** may be configured to accept monetary credits from casino patrons **262** and/or to dispense monetary credits to casino patrons **262** via cash, a credit or debit card, via a wireless interface (e.g., via a wireless payment app), via tickets, etc. According to some examples, the kiosks **260** may be configured to accept monetary credits from a casino patron and to provide a corresponding amount of monetary credits to a mobile gaming device **256** for wagering purposes, e.g., via a wireless link such as a near-field communications link. In some such examples, when a casino patron **262** is ready to cash out, the casino patron **262** may select a cash out option provided by a mobile gaming device **256**, which may include a real button or a virtual button (e.g., a button provided via a graphical user interface) in some instances. In some such examples, the mobile gaming device **256** may send a “cash out” signal to a kiosk **260** via a wireless link in response to receiving a “cash out” indication from a casino patron. The kiosk **260** may provide monetary credits to the casino patron **262** corresponding to the “cash out” signal, which may be in the form of cash, a credit ticket, a credit transmitted to a financial account corresponding to the casino patron, etc.

[0068] In some implementations, a cash-in process and/or a cash-out process may be facilitated by the TITO system server **108**. For example, the TITO system server **108** may control, or at least authorize, ticket-in and ticket-out transactions that involve a mobile gaming device **256** and/or a kiosk **260**.

[0069] Some mobile gaming devices **256** may be configured for receiving and/or transmitting player loyalty information. For example, some mobile gaming devices **256** may be configured for wireless communication with the player tracking system server **110**. Some mobile gaming devices **256** may be configured for receiving and/or transmitting player loyalty information via wireless communication with a patron's player loyalty card, a patron's smartphone, etc.

[0070] According to some implementations, a mobile gaming device **256** may be configured to provide safeguards that prevent the mobile gaming device **256** from being used by an unauthorized person. For example, some mobile gaming devices **256** may include one or more biometric sensors and may be configured to receive input via the biometric sensor(s) to verify the identity of an authorized patron. Some mobile gaming devices **256** may be configured to function only within a predetermined or configurable area, such as a casino gaming area.

[0071] FIG. 2C is a diagram that shows examples of components of a system for providing online gaming according to some aspects of the present disclosure. As with other figures presented in this disclosure, the numbers, types and arrangements of gaming devices shown in FIG. 2C are merely shown by way of example. In this example, various gaming devices, including but not limited to end user devices (EUDs) **264a**, **264b** and **264c** are capable of communication via one or more networks **417**. The networks **417** may, for example, include one or more cellular telephone networks, the Internet, etc. In this example, the EUDs **264a** and **264b** are mobile devices: according to this example the EUD **264a** is a tablet device and the EUD **264b** is a smart phone. In this implementation, the EUD **264c** is a laptop computer that is located within a residence **266** at the time depicted in FIG. 2C. Accordingly, in this example the hardware of EUDs is not specifically configured for online gaming, although each EUD is configured with software for online gaming. For example, each EUD may be configured with a web browser. Other implementations may include other types of EUD, some of which may be specifically configured for online gaming.

[0072] In this example, a gaming data center **276** includes various devices that are configured to provide online wagering games via the networks **417**. The gaming data center **276** is capable of communication with the networks **417** via the gateway **272**. In this example, switches **278** and routers **280** are configured to provide network connectivity for devices of the gaming data center **276**, including storage devices **282a**, servers **284a** and one or more workstations **570a**. The servers **284a** may, for example, be configured to provide access to a library of games for online game play. In some examples, code for executing at least some of the games may initially be stored on one or

more of the storage devices **282a**. The code may be subsequently loaded onto a server **284a** after selection by a player via an EUD and communication of that selection from the EUD via the networks **417**. The server **284a** onto which code for the selected game has been loaded may provide the game according to selections made by a player and indicated via the player's EUD. In other examples, code for executing at least some of the games may initially be stored on one or more of the servers **284a**. Although only one gaming data center **276** is shown in FIG. 2C, some implementations may include multiple gaming data centers **276**.

[0073] In this example, a financial institution data center **270** is also configured for communication via the networks **417**. Here, the financial institution data center **270** includes servers **284b**, storage devices **282b**, and one or more workstations **286b**. According to this example, the financial institution data center **270** is configured to maintain financial accounts, such as checking accounts, savings accounts, loan accounts, etc. In some implementations one or more of the authorized users **274a-274c** may maintain at least one financial account with the financial institution that is serviced via the financial institution data center **270**.

[0074] According to some implementations, the gaming data center **276** may be configured to provide online wagering games in which money may be won or lost. According to some such implementations, one or more of the servers **284a** may be configured to monitor player credit balances, which may be expressed in game credits, in currency units, or in any other appropriate manner. In some implementations, the server(s) **284a** may be configured to obtain financial credits from and/or provide financial credits to one or more financial institutions, according to a player's "cash in" selections, wagering game results and a player's "cash out" instructions. According to some such implementations, the server(s) **284a** may be configured to electronically credit or debit the account of a player that is maintained by a financial institution, e.g., an account that is maintained via the financial institution data center **270**. The server(s) **284a** may, in some examples, be configured to maintain an audit record of such transactions.

[0075] In some alternative implementations, the gaming data center **276** may be configured to provide online wagering games for which credits may not be exchanged for cash or the equivalent. In some such examples, players may purchase game credits for online game play, but may not "cash out" for monetary credit after a gaming session. Moreover, although the financial institution data center **270** and the gaming data center **276** include their own servers and storage devices in this example, in some examples the financial institution data center **270** and/or the gaming data center **276** may use offsite "cloud-based" servers and/or storage devices. In some alternative examples, the financial institution data center **270** and/or the gaming data center **276** may rely entirely on cloud-based servers.

[0076] One or more types of devices in the gaming data center **276** (or elsewhere) may be capable of executing middleware, e.g., for data management and/or device communication. Authentication information, player tracking information, etc., including but not limited to information obtained by EUDs **264** and/or other information regarding authorized users of EUDs **264** (including but not limited to the authorized users **274a-274c**), may be stored on storage devices **282** and/or servers **284**. Other game-related information and/or software, such as information and/or software relating to leaderboards, players currently playing a game, game themes, game-related promotions, game competitions, etc., also may be stored on storage devices **282** and/or servers **284**. In some implementations, some such game-related software may be available as "apps" and may be downloadable (e.g., from the gaming data center **276**) by authorized users.

[0077] In some examples, authorized users and/or entities (such as representatives of gaming regulatory authorities) may obtain gaming-related information via the gaming data center **276**. One or more other devices (such as EUDs **264** or devices of the gaming data center **276**) may act as intermediaries for such data feeds. Such devices may, for example, be capable of applying data filtering algorithms, executing data summary and/or analysis software, etc. In some implementations, data filtering, summary and/or analysis software may be available as "apps" and

downloadable by authorized users.

[0078] FIG. 3 illustrates, in block diagram form, an implementation of a game processing architecture **300** that implements a game processing pipeline for the play of a game in accordance with various implementations described herein. As shown in FIG. 3, the gaming processing pipeline starts with having a UI system **302** receive one or more player inputs for the game instance. Based on the player input(s), the UI system **302** generates and sends one or more RNG calls to a game processing backend system **314**. Game processing backend system **314** then processes the RNG calls with RNG engine **316** to generate one or more RNG outcomes. The RNG outcomes are then sent to the RNG conversion engine **320** to generate one or more game outcomes for the UI system **302** to display to a player. The game processing architecture **300** can implement the game processing pipeline using a gaming device, such as gaming devices **104A-104X** and **200** shown in FIGS. 1 and 2, respectively. Alternatively, portions of the gaming processing architecture **300** can implement the game processing pipeline using a gaming device and one or more remote gaming devices, such as central determination gaming system server **106** shown in FIG. 1.

[0079] The UI system **302** includes one or more UIs that a player can interact with. The UI system **302** could include one or more game play UIs **304**, one or more bonus game play UIs **308**, and one or more multiplayer UIs **312**, where each UI type includes one or more mechanical UIs and/or graphical UIs (GUIs). In other words, game play UI **304**, bonus game play UI **308**, and the multiplayer UI **312** may utilize a variety of UI elements, such as mechanical UI elements (e.g., physical “spin” button or mechanical reels) and/or GUI elements (e.g., virtual reels shown on a video display or a virtual button deck) to receive player inputs and/or present game play to a player. Using FIG. 3 as an example, the different UI elements are shown as game play UI elements **306A-306N** and bonus game play UI elements **310A-310N**.

[0080] The game play UI **304** represents a UI that a player typically interfaces with for a base game. During a game instance of a base game, the game play UI elements **306A-306N** (e.g., GUI elements depicting one or more virtual reels) are shown and/or made available to a user. In a subsequent game instance, the UI system **302** could transition out of the base game to one or more bonus games. The bonus game play UI **308** represents a UI that utilizes bonus game play UI elements **310A-310N** for a player to interact with and/or view during a bonus game. In one or more implementations, at least some of the game play UI element **306A-306N** are similar to the bonus game play UI elements **310A-310N**. In other implementations, the game play UI element **306A-306N** can differ from the bonus game play UI elements **310A-310N**.

[0081] FIG. 3 also illustrates that UI system **302** could include a multiplayer UI **312** purposed for game play that differs or is separate from the typical base game. For example, multiplayer UI **312** could be set up to receive player inputs and/or presents game play information relating to a tournament mode. When a gaming device transitions from a primary game mode that presents the base game to a tournament mode, a single gaming device is linked and synchronized to other gaming devices to generate a tournament outcome. For example, multiple RNG engines **316** corresponding to each gaming device could be collectively linked to determine a tournament outcome. To enhance a player's gaming experience, tournament mode can modify and synchronize sound, music, reel spin speed, and/or other operations of the gaming devices according to the tournament game play. After tournament game play ends, operators can switch back the gaming device from tournament mode to a primary game mode to present the base game. Although FIG. 3 does not explicitly depict that multiplayer UI **312** includes UI elements, multiplayer UI **312** could also include one or more multiplayer UI elements.

[0082] Based on the player inputs, the UI system **302** could generate RNG calls to a game processing backend system **314**. As an example, the UI system **302** could use one or more application programming interfaces (APIs) to generate the RNG calls. To process the RNG calls, the RNG engine **316** could utilize gaming RNG **318** and/or non-gaming RNGs **319A-319N**. Gaming RNG **318** could corresponds to RNG **212** or hardware RNG **244** shown in FIG. 2A. As

previously discussed with reference to FIG. 2A, gaming RNG **318** often performs specialized and non-generic operations that comply with regulatory and/or game requirements. For example, because of regulation requirements, gaming RNG **318** could correspond to RNG **212** by being a cryptographic RNG or pseudorandom number generator (PRNG) (e.g., Fortuna PRNG) that securely produces random numbers for one or more game features. To securely generate random numbers, gaming RNG **318** could collect random data from various sources of entropy, such as from an operating system (OS) and/or a hardware RNG (e.g., hardware RNG **244** shown in FIG. 2A). Alternatively, non-gaming RNGs **319A-319N** may not be cryptographically secure and/or be computationally less expensive. Non-gaming RNGs **319A-319N** can, thus, be used to generate outcomes for non-gaming purposes. As an example, non-gaming RNGs **319A-319N** can generate random numbers for generating random messages that appear on the gaming device.

[0083] The RNG conversion engine **320** processes each RNG outcome from RNG engine **316** and converts the RNG outcome to a UI outcome that is feedback to the UI system **302**. With reference to FIG. 2A, RNG conversion engine **320** corresponds to RNG conversion engine **210** used for game play. As previously described, RNG conversion engine **320** translates the RNG outcome from the RNG **212** to a game outcome presented to a player. RNG conversion engine **320** utilizes one or more lookup tables **322A-322N** to regulate a prize payout amount for each RNG outcome and how often the gaming device pays out the derived prize payout amounts. In one example, the RNG conversion engine **320** could utilize one lookup table to map the RNG outcome to a game outcome displayed to a player and a second lookup table as a pay table for determining the prize payout amount for each game outcome. In this example, the mapping between the RNG outcome and the game outcome controls the frequency in hitting certain prize payout amounts. Different lookup tables could be utilized depending on the different game modes, for example, a base game versus a bonus game.

[0084] After generating the UI outcome, the game processing backend system **314** sends the UI outcome to the UI system **302**. Examples of UI outcomes are symbols to display on a video reel or reel stops for a mechanical reel. In one example, if the UI outcome is for a base game, the UI system **302** updates one or more game play UI elements **306A-306N**, such as symbols, for the game play UI **304**. In another example, if the UI outcome is for a bonus game, the UI system could update one or more bonus game play UI elements **310A-310N** (e.g., symbols) for the bonus game play UI **308**. In response to updating the appropriate UI, the player may subsequently provide additional player inputs to initiate a subsequent game instance that progresses through the game processing pipeline.

[0085] FIG. 4A is a perspective view of a portion of an EGM **104** that includes a mechanical enclosure assembly **400** for securely housing a currency acceptor, such as a bill validator or bank notes acceptor **124** (shown in FIG. 4B), a ticket printer **126**, and/or any other currency or ticket printing and/or accepting device. FIG. 4B is a perspective view of enclosure assembly **400** shown in FIG. 4A. FIG. 5 is a front view of enclosure assembly **400**. FIG. 6 is an exploded view of enclosure assembly **400**.

[0086] Referring to FIG. 4A, EGM **104** includes a cabinet **401** and a button deck door assembly **403** coupled to cabinet **401**. Button deck door assembly **403** is moveable relative to cabinet **401** (e.g., rotated outwards from cabinet **401** in the example embodiment) between a closed position, as shown in FIG. 4A, and an open position (not shown). When in the open position, an interior of cabinet **401** is accessible to an operator. Button deck door assembly **403** includes a button deck **405**, a ticket out printer cover **407**, a button deck display **409** and a BNA cover **411**. Enclosure assembly **400** includes a door **428** for providing access to an interior portion **420** (shown in FIG. 5). Enclosure assembly **400** does not rotate with button deck door assembly **403**. When button deck door assembly **403** is in the closed position, as shown in FIG. 4A, a slight gap **413** is defined between BNA cover **411** and door **428**. When button deck door assembly **403** is rotated to the open position, BNA cover **411** is moved from enclosure assembly **400**, exposing an enclosure opening

419 (shown in FIG. 5).

[0087] Enclosure assembly **400** may be removably positioned within and/or attached to EGM **104**, such as within an internal channel or receptacle of a EGM **104**, or attached at a side of cabinet **401**, as shown in the example embodiment. Referring to FIG. 4B, during operation, the BNA **124** may receive paper currency and/or paper tickets through an insert in BNA cover **411** (shown in FIG. 4A) at a bill scanner **415** of BNA **124**, which may be stacked and/or stored within a bill cassette or cashbox **421** of the BNA **124**. Periodically, a service technician may remove BNA **124** from EGM **104** to access and remove the paper currency and/or tickets stacked and stored within cashbox **421**. Thereafter, the technician may replace the empty BNA **124** within the EGM **104** for continued use.

[0088] As a result of the money handling functions performed by BNA **124**, in many embodiments, it is important that BNA **124** remain secure within the EGM **104**. For example, it may be desirable to ensure that BNA **124** is secure against tampering, theft, and/or any other unwanted human interaction. To secure BNA **124**, mechanical enclosure assembly **400** may thus be provided, which as described herein may be arranged to receive and securely contain BNA **124**, while also allowing removal of BNA **124** from the EGM **104** when authorized and appropriate, such as, for example, by an authorized technician. Further, as described herein, enclosure assembly **400** may also facilitate quick, quite, and safe removal of BNA **124** from the EGM **104** by an authorized technician.

[0089] Accordingly, in the example embodiment, enclosure assembly **400** includes a housing **402** and a door assembly **404** hingeably coupled to housing **402** at a pivot point **406**, such as near a lower portion **408** of housing **402**. As a result, door assembly **404** is generally capable of arcuate or angular motion (e.g., opening and closing) relative to housing **402** about pivot point **406**.

[0090] Referring to FIGS. 4B-6, collectively, in the example embodiment, housing **402** includes a first side wall **410**, a second side wall **412** spaced apart from first side wall **410** by a width, W, and a connecting portion **414** extending between and mechanically coupling first side wall **410** and second side wall **412**. As shown, housing **402** may be generally polyhedral and shaped to receive and contain BNA **124**, as described herein. As a result, although a polyhedral shaping that includes a sloping portion **416** is illustrated, it will be appreciated that housing **402** may include any other shape adapted to receive BNA **124** and/or another ticket and/or currency accepting and/or printing device. For example, as shown, and in at least one embodiment, housing **402** is generally “saddlebag” shaped.

[0091] In the example embodiment, housing **402** also defines an exterior portion **418**, an interior portion **420**, and an opening **422**. In at least some embodiments, exterior portion **418** may engage, such as in a friction or pressure fit, within a recess, a channel, and/or another compartment or receptacle of EGM **104**. In other embodiments, exterior portion **418** may not be friction or pressure fit within EGM **104** but may nonetheless be positioned within a recess or channel of EGM **104**. As described herein, interior portion **420** may receive and contain BNA **124**.

[0092] In some embodiments, enclosure assembly **400** includes at least one engagement bracket **424** coupled to side portion **412** and extending partway into interior portion **420** of housing **402**. Engagement bracket **424** may engage a prong latch **456** (shown in FIG. 8) or another corresponding engagement surface of door assembly **404**. In the example embodiment, engagement bracket **424** is a stainless-steel flat spring. In some embodiments, one or more additional engagement surfaces (e.g., flat springs) may be included as well. For example, in at least one implementation, a second flat spring may be included on side portion **412** opposite flat spring **424**. Further, although flat springs are described herein, it will be appreciated that any other suitable engagement mechanism, such as any pressure-fit and/or friction-fit locking or engagement mechanism, may also be used to secure BNA **124** within housing **402**.

[0093] Accordingly, housing **402** may be sized and shaped to receive BNA **124** and/or, as described above, to accommodate removal and replacement of BNA **124**, such as by a technician for removal of paper currency from BNA **124** during operation and/or on a periodic basis. To facilitate removal and replacement of BNA **124**, housing **402** includes opening **422**, the dimensions of which may be

sufficient to permit placement of BNA within housing **402** as well as removal of BNA from housing **402**.

[0094] As described above, enclosure assembly **400** also includes door assembly **404** hingeably coupled to housing **402** at pivot point **406**, such as near lower portion **408** of housing **402**. Also, as described above, door assembly **404** is generally capable of arcuate or angular motion (e.g., opening and closing) relative to housing **402** about pivot point **406**.

[0095] More particularly, door assembly **404** includes a torque damper assembly **426** and a door panel **428** coupled to the torque damper assembly **426**. In at least some implementations, torque damper assembly **426** may be generally configured to facilitate, and dampen, motion of door assembly **404** relative to opening **422**, such that door assembly **404** is capable of opening and closing to provide access to interior portion **420** of housing **402** when door assembly **404** is open, and when door assembly **404** is closed, to at least partially occlude opening **422** and secure BNA **124** within housing **402**.

[0096] Torque damper assembly **426** is shown, in additional detail, with reference to FIG. 7. Accordingly, with primary reference to FIG. 7, torque damper assembly **426** may include, in at least the example implementation, a platform **430**, a torque damper **432** rotatably coupled to platform **430** at a first pivot point **434**, a first arm **436** fixedly coupled to torque damper **432**, and a second arm **438** rotatably coupled to platform **430** at a second pivot point **440**. Although in the example implementation, second arm **438** is not coupled to a respective torque damper, in at least some embodiments, second arm **438** may also or alternatively be coupled to a torque damper (e.g., torque damper assembly **426** may include one or more torque dampers **432**).

[0097] In at least some embodiments, torque damper **432** may include any suitable torque damping mechanism, such as one or more springs capable of applying a biasing force against rotation of door assembly **404** (and/or door panel **428**) away from opening **422** (e.g., a biasing force that generally opposes opening of door assembly **404**). In the example embodiment, torque damper **432** is a hydraulic damper. Likewise, in some embodiments, torque damper **432** may also assist closure of door assembly, such as by countering at least a portion of the weight of door assembly **404**.

[0098] Moreover, in the example implementation, torque damper **432** may facilitate a generally unopposed or undamped (and thus rapid) opening of door assembly **404** through approximately two-thirds of the range of motion of door assembly **404**, from a closed position to an open position. During approximately the last one-third of the range of motion of door assembly, torque damper **432** may provide a biasing or dampening force, as described above, that substantially opposes and counters at least a portion of the weight of door assembly to dampen and slow door assembly **404** in the last one-third of the range, as door assembly **404** reaches the open position. Other ranges are also contemplated by and within the scope of the present disclosure. For example, torque damper **432** may supply a biasing force in a range of about fifty percent (e.g., half-way open), and/or any other suitable range.

[0099] FIG. 8 is a front view of enclosure assembly **400**, with door panel **428** removed. In the example embodiment, enclosure assembly **400** further includes a lock assembly **442** for securely locking door assembly **404** in a closed position on housing **402**. Lock assembly **442** includes a lock frame **444**, engagement bracket **446**, bore frame **448** and first and second latch blocks **450**, **452**. Lock frame **444** is coupled to door panel **428** (shown in FIG. 6) and supports latch blocks **450**, **452** therein. First and second latch blocks **450**, **452** are each slideably moveable relative to lock frame **444** between a locked position, as shown in FIG. 8, and an unlocked position (not shown). In the locked position, first and second latch blocks **450**, **452** are selectively engaged with bore frame **448** and engagement bracket **446**, respectively.

[0100] First and second latch blocks **450**, **452** each include an upper or first prong latch **454**, **456** and a lower or second prong latch **458**, **460**. In the example embodiment, upper and lower prong latches **454**, **458** of first latch block **450** each engage bore frame **448** while only upper prong latch **456** of second latch block **452** engages engagement bracket **446**. Additionally or alternatively to

engagement bracket **446**, enclosure assembly **400** in other embodiments may include a second engagement bracket (not shown) positioned in alignment with lower prong latch **460** of second latch block **452** and that is engaged by lower prong latch **460** when in the locked position. In further embodiments, enclosure assembly **400** may include any number of engagement brackets similar to engagement bracket **446** that are engaged with any one of prong latches **454-460**. For example, and without limitation, in some embodiments, enclosure assembly **400** does not include bore frame **448** and instead includes four engagement brackets, each corresponding to one of prong latches **454-460**.

[0101] An actuator **462** is connected to latch blocks **450**, **452** and may be turned or rotated, such as with a physical key (not shown), to extend and retract latch blocks **450**, **452** into and out of engagement with engagement bracket **446** and bore frame **448**. As shown in FIG. **8**, first latch block **450** and second latch block **452** are each in a locked position. To unlock lock assembly **442** when opening door panel **428** (shown in FIG. **4B**), actuator **462** is rotated, which actuates a cam connector (not shown), moving first latch block **450** in the direction A and out of engagement with bore frame **448**. Rotation of actuator **462** further moves second latch block **452** in the direction B, substantially opposite direction A, and out of engagement with engagement bracket **446**.

[0102] In the example embodiment, bore frame **448** is slidable and/translatable in the directions, "C" and "D," such that bore frame **448** can be translated vertically relative to a side portion **464** of BNA **124**. As a result, in at least one implementation, bore frame **448** may be translated in the direction "C" to permit removal of BNA **124** from enclosure assembly **400**. Likewise, to retain BNA **124** within enclosure assembly **400**, bore frame **448** may be translated in the direction "D," which causes bore frame **448** to physically impede removal of BNA **124** from enclosure assembly **400**. A fastener **466** can be adjusted to permit translation of bore frame **448** as desired.

[0103] FIG. **9** is a perspective view of a portion of enclosure assembly showing first latch block **450** engaged with bore frame **448**. In the example embodiment, bore frame **448** includes a pair of side walls **468** (one shown in FIG. **9**) extending inward from first side wall **410** of housing **402** and an end wall **470** extending between side walls **468**. End wall **470** defines an upper opening **472** and a lower opening **474**. Upper and lower prong latches **454**, **458** of first latch block **450** each include a first prong **478** extending into a respective one of the openings **472**, **474**, and a second prong **480** that contacts bore frame side wall **468**. Accordingly, when first prongs **478** are received in openings **472**, **474** of bore frame **448** lateral movement of door panel **428** (shown in FIG. **4B**) relative to first side wall **410** is restricted.

[0104] FIG. **10** is a perspective view of another portion of enclosure assembly **400** showing second latch block **452** engaged with engagement bracket **424**. FIG. **11** is a top view of the portion of enclosure assembly **400** shown in FIG. **9**.

[0105] Referring to FIG. **9**, second latch block **452** is substantially the same as first latch block **450**, in that upper prong latch **456** and lower prong latch **460** each include a pair of first and second prongs **478**, **480**. Engagement bracket **424** is attached to second side wall **412** and extends through an opening **482** (shown in FIG. **12**) defined in second side wall **412**. First prong **478** of upper prong latch **456** extends into a notch or slot **484** defined in engagement bracket **424**. Lock frame **444** includes a side surface **486** facing second side wall **412** and a projection **488** that extends outward from side surface **486** toward second side wall **412**.

[0106] Referring to FIG. **10**, engagement bracket **424** is spaced from and out of contact with projection **488** of lock frame **444**. Engagement bracket **424** is configured to flex in response to a force exerted on engagement bracket **424** to contact projection **488** and resist opening of door panel **428** or more specifically, resist movement of lock assembly **442** relative to second side wall **412**.

[0107] In the example embodiment, engagement bracket **424** extends between a first end **490** attached to second side wall **412** by a fastener **492**, to a second free distal end **494**. Engagement bracket **424** includes a first surface **496** extending from first end **490** and a second surface **498** that extends from first surface **496** to second end **494**. First surface **496** and second surface **498** have an

offset orientation from one another, and are specifically obliquely oriented to one another as shown in FIG. 11. A bend point **500** of engagement bracket **424** is defined at the intersection of first surface **496** and second surface **498**. Notch **484** (shown in FIG. 9) is defined within first surface **496** and second surface **498**, though in other embodiments, notch **484** may be defined at any position along engagement bracket **424**.

[0108] As shown in FIG. 11, during normal operations, engagement bracket **424** is out of contact with projection. In at least some embodiments, if enclosure assembly **400** is tampered with, such as by a person attempting to break into enclosure assembly **400**, any outward or other “pulling” force exerted on door assembly **404** (i.e., in the direction E) will tend to drive bend point **500** and/or first and second surfaces **496**, **498** of engagement bracket **424** into frictional engagement with projection, which may in turn, increase the structural rigidity and/or strength of enclosure **400**, as door assembly **404** is wedged or driven into tighter contact with enclosure **400**. Specifically, with first prong **478** extended through notch **484**, movement of first prong **478** in the direction E causes engagement bracket to bend at bend point **500**, moving first surface **496** and second surface **498** inward of second side wall **412** and into contact with projection **488** of lock frame **444**.

Accordingly, attempts to tamper with enclosure assembly **400** may in fact result in structural reinforcement of enclosure assembly **400** rather than any weakening of assembly **400**.

[0109] FIG. 12 is a side perspective view of enclosure assembly **400**, with portions of door assembly **404** and lock assembly **442** removed to reveal a switch **502**. FIG. 13 is a top perspective view of enclosure assembly **400**, illustrating first and second latch blocks **450**, **452**.

[0110] Referring to FIG. 12, in the example embodiment enclosure assembly **400** further includes a switch **502** coupled to first side wall **410**. Switch **502** is configured to detect whether door panel **428** is in the open or closed positions. Switch **502** includes a compressible push button **504** that is compressed when door panel **428** is in the closed position and extended when door panel **428** is in the open position. Switch **502** is coupled in communication to a controller, EGM game controller **202** shown in FIG. 2A, within cabinet **401** (shown in FIG. 4A). In some embodiments, when switch **502** detects that door panel **428** is open, an alarm or alert may be triggered on EGM **104** or transmitted to a central server for notifying a technician.

[0111] Referring to FIG. 13, door panel **428** further includes at least one end flange **506** that extends around at least a portion of one of the side walls **410**, **412** of housing **402**. Specifically, in the example embodiment, first side wall **410** includes a hook wall **508** that extends inward of end flange **506** of door panel **428**. As a result, hook wall **508** and end flange **506** define a winding gap **510** between door assembly **404** and first side wall **410**, obstructing linear insertion of any tampering devices, such as a lock pick, through the winding gap **510**.

[0112] FIGS. 14 and 15 show a gap closing assembly **512** of enclosure assembly **400**. Specifically, FIG. 14 shows gap closing assembly **512** in a raised position when door panel **428** is closed and FIG. 15 shows gap closing assembly **512** in a lowered position when door panel **428**.

[0113] Gap closing assembly **512** is configured to extend across gap **413** (shown in FIG. 4A) in the raised position when door panel **428** is closed to obstruct any intrusions into enclosure assembly through gap **413**. Additionally, gap closing assembly **512** is at least partially moveable relative to door panel **428** such that door panel **428** may be opened without obstruction between gap closing assembly **512** and BNA cover **411** (shown in FIG. 4A). Specifically, when gap closing assembly **512** is in the raised position, as shown in FIG. 14, a rotatable cover **514** extends vertically across gap **413** and at least partially within an interior of BNA cover **411** (shown in FIG. 4A) to close off gap **413**. When door panel **428** is opened, rotatable cover **514** of gap closing assembly **512** is passively rotated to the lowered position (shown in FIG. 15) to clear BNA cover **411** and allow for opening of door panel **428**.

[0114] Referring to FIG. 14, gap closing assembly **512** includes a shaft **516**, rotatable cover **514**, a back stop **518**, and a biasing element **520**. Shaft **516** is attached to door panel **428** and is supported on door panel **428** by an arm **522**, extending inward from an interior surface **521** of door panel **428**.

Shaft **516** is positioned within arm **522** and is rotatable relative thereto about a rotational axis R.sub.1. In other embodiments, shaft **516** does not rotate and rotatable cover **514** may be rotatably mounted on shaft **516**.

[0115] Rotatable cover **514** includes a flap portion **524**, a ring portion **526**, and a pair of stop feet **528**. Ring portion **526** wraps around shaft **516** to attach rotatable cover **514** to shaft **516**. Flap portion **524** extends radially outward from ring portion **526**. Stop feet **528** extend in a generally opposite direction from ring portion **526** as flap portion **524**. In the example embodiment, rotatable cover includes two stop feet **528** (shown in FIG. 15), though any suitable number of stop feet **528** may be included.

[0116] Biasing element **520** is attached to shaft **516** and engages flap portion **524** to bias flap portion **524** inward of door panel **428** (i.e., in a clockwise direction on shaft **516** as shown in FIG. 2. In the example embodiment, biasing element **520** is a torsion spring, though other suitable biasing elements may also be used. A back stop **518** is attached to second side wall **412** and extends inward from second side wall **412** to contact flap portion **524** when door panel **428** is in the closed position. A substantially identical back stop is also attached to first side wall **410** of enclosure assembly. Back stops **518** hold flap portion **524** in the raised position against a biasing force of biasing element **520**.

[0117] As door panel **428** is opened, shaft **516** and rotatable cover **514** are carried with door panel **428** away from back stops **518** and biasing element **520** biases rotatable cover **514** to rotate with shaft **516** to the lowered position, as shown in FIG. 15. In the example embodiment, flap portion **524** rotates approximately 60 degrees between the raised position and lowered position. In the lowered position, flap portion **524** does not overlap BNA cover **411** and door panel **428** may be fully opened without obstruction.

[0118] Referring to FIG. 15, in the lowered position, stop feet **528** each contact at least a portion of door panel **428**, providing a stop against further rotation of rotatable cover **514**. When door panel **428** is raised to again close door panel **428**, flap portion **524** will clear BNA cover **411** (FIG. 4A) and contact back stop **518**. Engagement of flap portion **524** with back stop **518** overcomes biasing force provided by biasing element **520** on flap portion **524** and causes flap portion to rotate outwards (i.e., counter-clockwise in FIG. 15) to the raised position, as shown in FIG. 14.

[0119] Exemplary technical effects of the methods, systems, and apparatus described herein include at least one of: (a) providing a secure locking of a BNA within an enclosure of an electronic gaming machine; (b) reduced size of a lock assembly of a BNA enclosure, as compared with at least some systems that use multiple bore frames mounted to an enclosure; (c) reduced overall damage to door panels by damping a portion of the angular motion of the door panel when opened; (d) reduced complexity in servicing, installation, and/or removal of BNA from an enclosure assembly; (e) reduced overall time required in servicing, installation, and/or removal of a BNA from an enclosure assembly; (f) improved tamper prevention, at least in part due to the mechanical strength of the enclosure assembly provided by engagement between the engagement bracket and the projection; (g) reduced accessible openings between enclosure assembly interior due at least in part, to the moveable gap cover assembly.

[0120] While the disclosure has been described with respect to the figures, it will be appreciated that many modifications and changes may be made by those skilled in the art without departing from the spirit of the disclosure. Any variation and derivation from the above description and figures are included in the scope of the present disclosure as defined by the claims.

[0121] This written description uses examples to disclose the disclosure, including the best mode, and also to enable any person skilled in the art to practice the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have

structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

Claims

1. An enclosure assembly for use with an electronic gaming machine (EGM), the enclosure assembly comprising: a housing sized to receive at least one accessory component of the EGM; a door panel coupled to the housing and moveable between an open position and a closed position, the door panel and the housing defining a gap therebetween when the door panel is in the closed position; a latch coupled to the door panel and positioned to engage a flexible engagement bracket on the housing for locking the door panel in the closed position; and a gap closing assembly coupled to the door panel and positioned to at least partially cover the gap when the door panel is in the closed position.
2. The enclosure assembly of claim 1, wherein the gap closing is rotatable relative to the door panel between a raised position in which the gap closing assembly is positioned to at least partially cover the gap, and a lowered position.
3. The enclosure assembly of claim 1, wherein the gap closing assembly comprises: a shaft rotatably coupled to the door panel; a rotatable cover coupled to the shaft, the rotatable cover being rotatable between a raised position in which the cover is positioned to at least partially cover the gap, and a lowered position; and a biasing element coupled to the shaft and biasing the rotatable cover to the lowered position.
4. The enclosure assembly of claim 3, wherein the rotatable cover obstructs opening of the door panel in the raised position and does not obstruct opening of the door panel when in the lowered position.
5. The enclosure assembly of claim 3, wherein the housing further includes a back stop that, when the door panel is closed, holds the rotatable cover in the raised position.
6. The enclosure assembly of claim 5, wherein, when the door panel is closed, the biasing element contacts a first side of the rotatable cover and the back stop contacts a second side of the rotatable cover.
7. The enclosure assembly of claim 1, wherein the latch causes the engagement bracket to flex into contact with at least a portion of the door panel in response to the door panel being pulled outward when the door panel is locked in the closed position.
8. The enclosure assembly of claim 1, wherein the engagement bracket defines a notch and wherein the latch extends into the notch when the door panel is locked to the housing.
9. The enclosure assembly of claim 1, wherein the engagement bracket further comprises a first surface extending from a first end of the engagement bracket and a second surface extending between the first surface and a second end of the engagement bracket, the second surface being obliquely oriented relative to the first surface.
10. The enclosure assembly of claim 9, wherein the engagement bracket defines a bend point between the first surface and the second surface, and wherein, when the door panel is locked to the housing, the bend point contacts a projection on the door panel in response to the door panel being pulled outward.
11. The enclosure assembly of claim 1 further comprising: a bore frame coupled to the housing and defining an opening therein; and an additional latch coupled to the door panel and extending into the opening of the bore frame, when the door panel is locked to the housing.
12. The enclosure assembly of claim 1 further comprising a torque damper configured to dampen at least a portion of an angular motion of the door panel relative to the housing, when the door panel is opened.
13. The enclosure assembly of claim 1, wherein the at least one accessory component includes one of a bank notes acceptor, a ticket printer, and a ticket acceptor.

14. An electronic gaming device comprising: a cabinet; a display coupled to the cabinet for displaying an electronic game thereon; and an enclosure assembly coupled to the cabinet, the enclosure assembly comprising: a housing sized to receive an accessory component of the gaming device; a door panel coupled to the housing and moveable between an open position and a closed position, the door panel and the housing defining a gap therebetween when the door panel is in the closed position; a latch coupled to the door panel and positioned to engage a flexible engagement bracket on the housing for locking the door panel in the closed position; and a gap closing assembly coupled to the door panel for at least partially covering the gap when the door panel is in the closed position.

15. The electronic gaming device of claim 14, wherein the gap closing is rotatable relative to the door panel between a raised position in which the gap closing assembly is positioned to at least partially cover the gap, and a lowered position.

16. The electronic gaming device of claim 14, wherein the gap closing assembly comprises: a shaft rotatably coupled to the door panel; a rotatable cover coupled to the shaft, the rotatable cover being rotatable between a raised position in which the cover is positioned to at least partially cover the gap, and a lowered position; and a biasing element coupled to the shaft and biasing the rotatable cover to the lowered position.

17. The electronic gaming device of claim 16, wherein the housing further includes a back stop that, when the door panel is closed, holds the rotatable cover in the raised position and wherein, when the door panel is closed, the biasing element contacts a first side of the rotatable cover and the back stop contacts a second side of the rotatable cover.

18. A method of securing an accessory component within an enclosure assembly of an electronic gaming device, the method comprising: positioning the accessory component within a housing of the enclosure assembly, wherein a flexible engagement bracket is coupled to the housing; moving a door panel from an open position to a closed position, the door panel being coupled to the housing, wherein the door panel and the housing define a gap therebetween when the door panel is in the closed position; moving a latch into engagement with the engagement bracket to lock the door in the closed position; and covering, at least partially, the gap with a gap closing assembly when the door panel is in the closed position, the gap closing assembly being coupled to the door panel.

19. The method of claim 18 further comprising rotating, passively, a cover of the gap closing assembly from a lowered position to a raised position as the door is moved to the closed position, wherein, in the raised position the cover is positioned to at least partially cover the gap.

20. The method of claim 19, wherein the gap closing assembly includes a biasing element biasing the rotatable cover to the lowered position, and wherein the cover obstructs opening of the door panel in the raised position and does not obstruct opening of the door panel when in the lowered position.
